Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Signed area

Meaning in my words How would you explain signed area to a friend?	**How can you tell?** What do you do with the area below the time axis when you find displacement?
Example or sketch Sketch a rectangle at −2 m/s lasting 3 s. What displacement does it show?	**Non-example and why** Someone labels the area −2 m/s. Which graph reading did they use?

Distance

Meaning in my words How would you explain distance to a friend?	**How can you tell?** A cart goes 3 m right and 2 m left. How much ground do its wheels cover?
Example or sketch Draw a trip that covers 6 m and finishes at its starting point.	**Non-example and why** A partner writes −5 m for distance. What does the minus sign tell you?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Component

Meaning in my words How would you explain component to a friend?	**How can you tell?** If east and north are positive, which component is negative for a southeast trip?
Example or sketch What two arrows could you use to build a displacement of 3 m east and 4 m north?	**Non-example and why** A classmate counts the diagonal and both parts as three separate trips. What is counted twice?

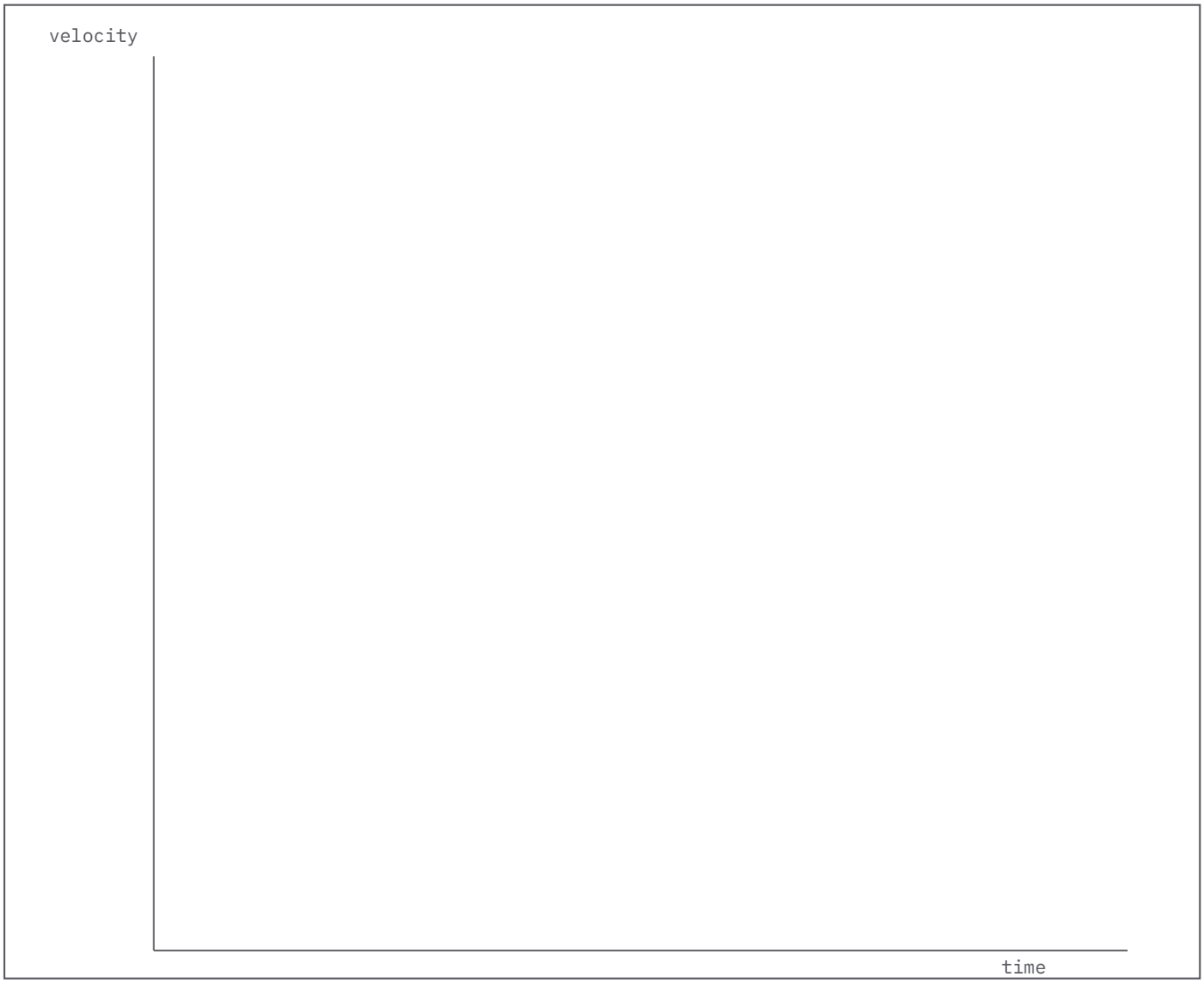
Angle θ

Meaning in my words When you label an arrow with an angle, what does that angle tell someone?	**How can you tell?** What extra information do you need when a vector is labeled “30 degrees”?
Example or sketch Show 30 degrees north of east on a sketch.	**Non-example and why** Does 30 degrees east of north point the same way as 30 degrees north of east?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw and shade opposite-side rectangles. Put m/s on the vertical and s on the horizontal axis.



My marks mean

On assignment: Signed area

SECTION E / BACK / FINAL MEETING ONLY Name _____

A moving camera follows a model at +3 km/s for 2 s, then −1 km/s for 2 s. The editor needs the end location and the total footage distance.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Tracking engineer**

Shade both parts of a velocity graph. Give the camera operator the final change from start, including direction.

☐ **Replay producer**

Storyboard the outbound and return shots. Tell the editor total distance and net movement; mark which part plays in reverse direction.

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

Editor's check: Turn the axis ninety degrees. Explain why transverse velocity needs its own graph.